

Determining the Effectiveness of a Membrane Separation Process

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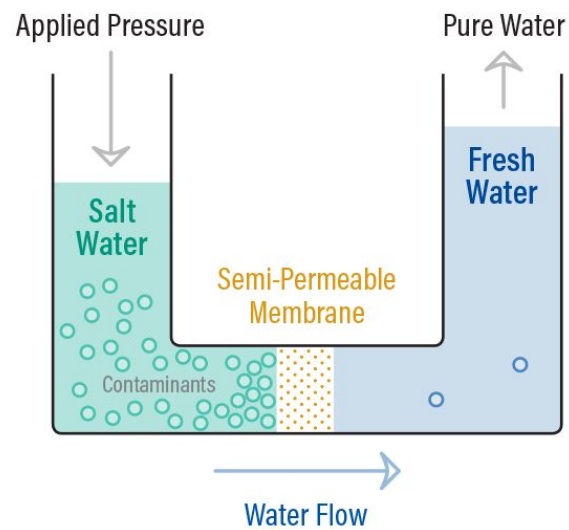
[horizonregional.com]

Overview



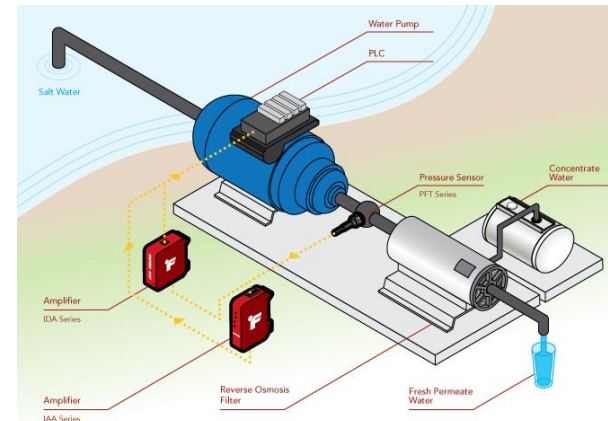
[samcotech.com]

What is reverse osmosis?



[puretecwater.com]

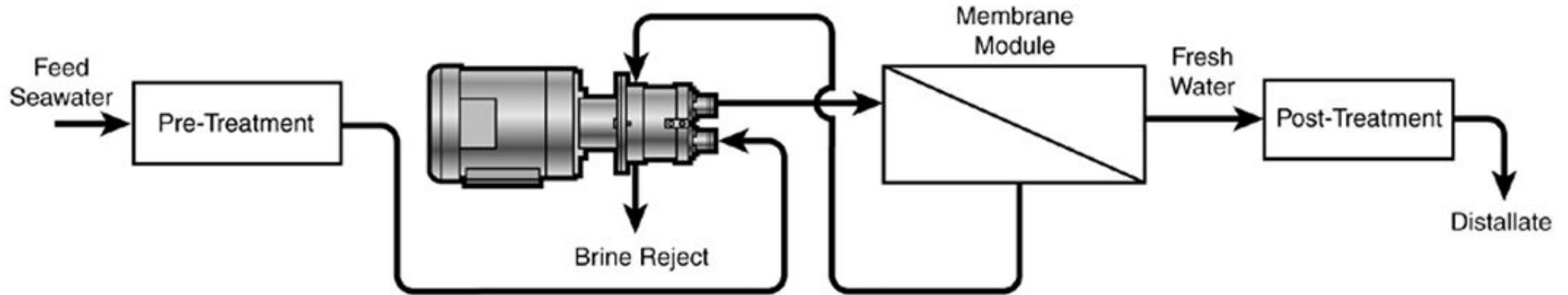
Motivation



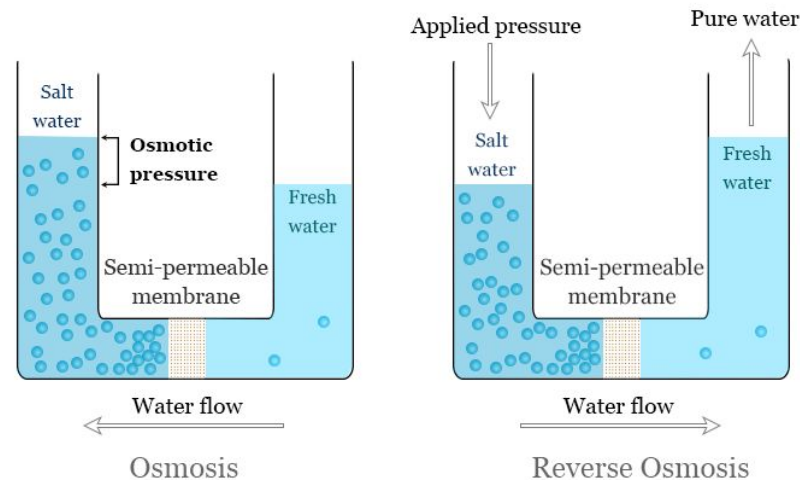
[futek.com]

Experiment and Results

Reverse osmosis separates dissolved solutes from water by forcing water through a selectively permeable membrane

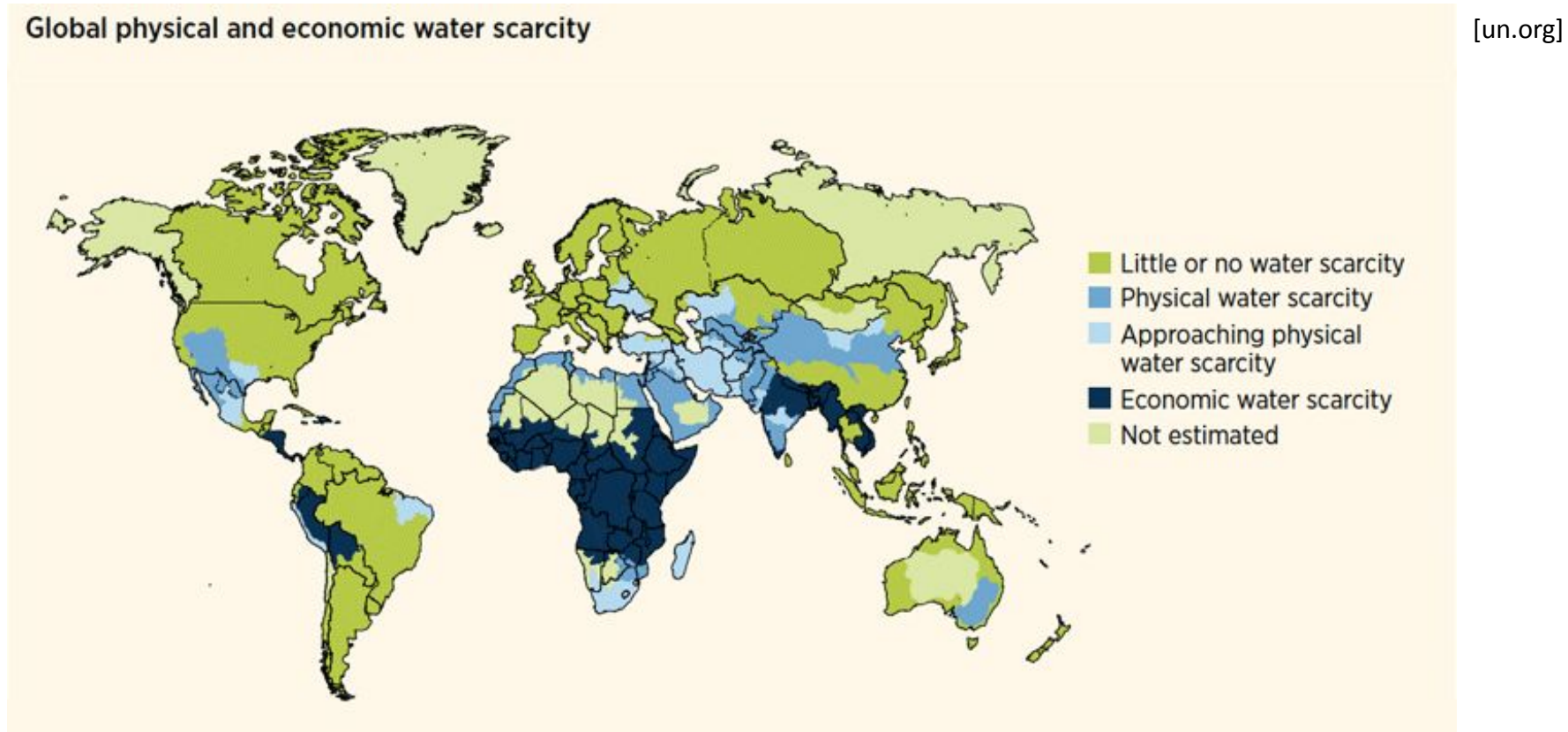


[nmwaterconference.nmwrri.nmsu.edu]



[priyamstudycentre.com]

There is a need to make RO a more cost-effective and energy-efficient process to purify seawater



Up to 4 billion people live under some water scarcity

There is a need to make RO a more cost-effective and energy-efficient process to purify seawater

Power consumption must grow to increase filtration performance

RO modules are used in series

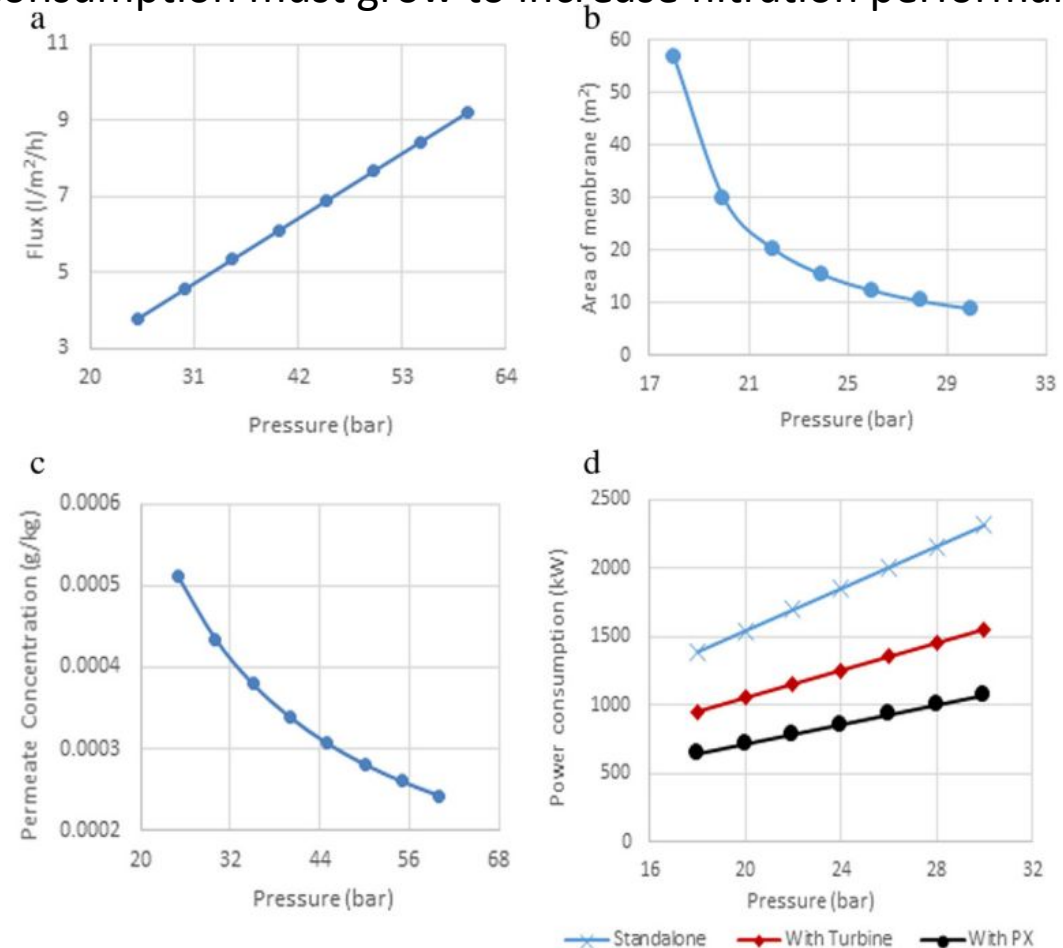


Fig. 6. Effect of pressure on permeate concentration, flux, area, ERT and PX power saving.

The Rejection Coefficient

Defined as the ratio between the concentration differential up and downstream of the membrane, and the feed concentration:

$$r = 1 - C_p / C_f$$

C_p

C_f

C_r



Recovery

Defined as the ratio between permeate flow rate and feed flow rate:

$$Y = Q_p / Q_f$$

Q_p

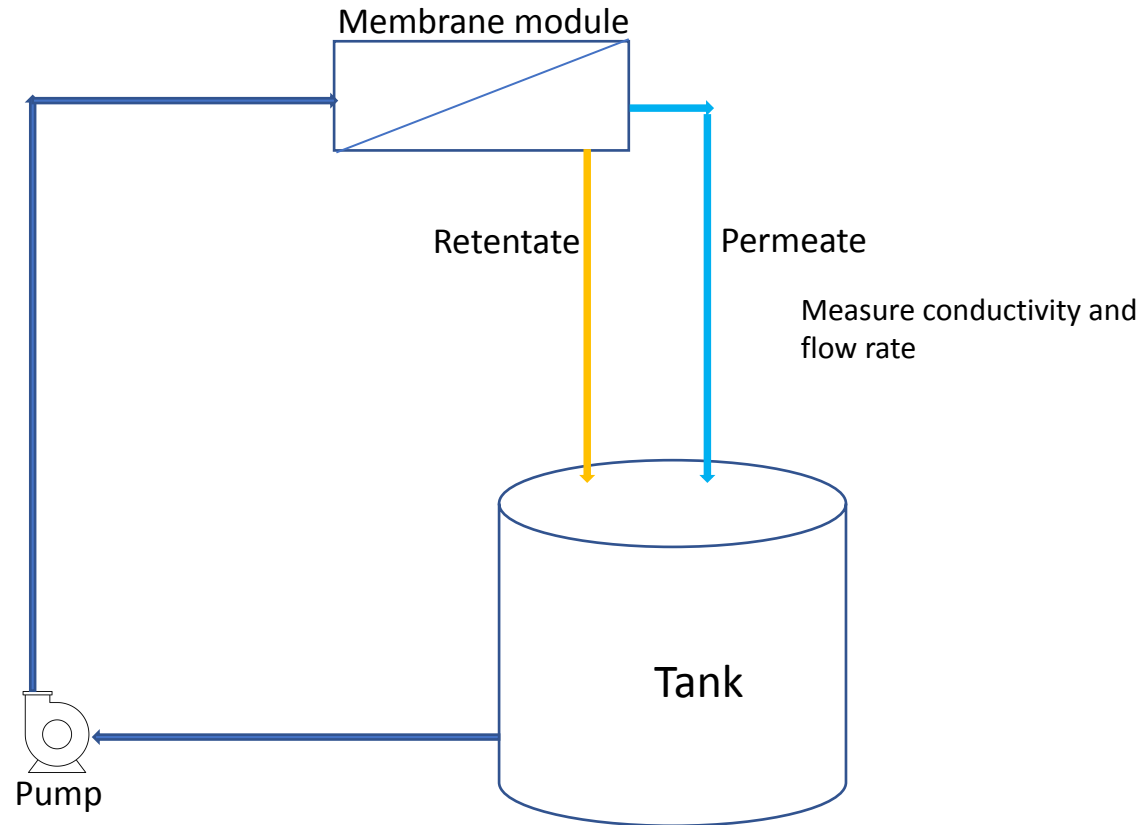


Q_f

Permeate and retentate concentration in a closed system was measured for varying feed NaCl concentrations and pressures

Performed trials
for 0.11 to 0.31%
NaCl content

Trials from 2 to 5
bar

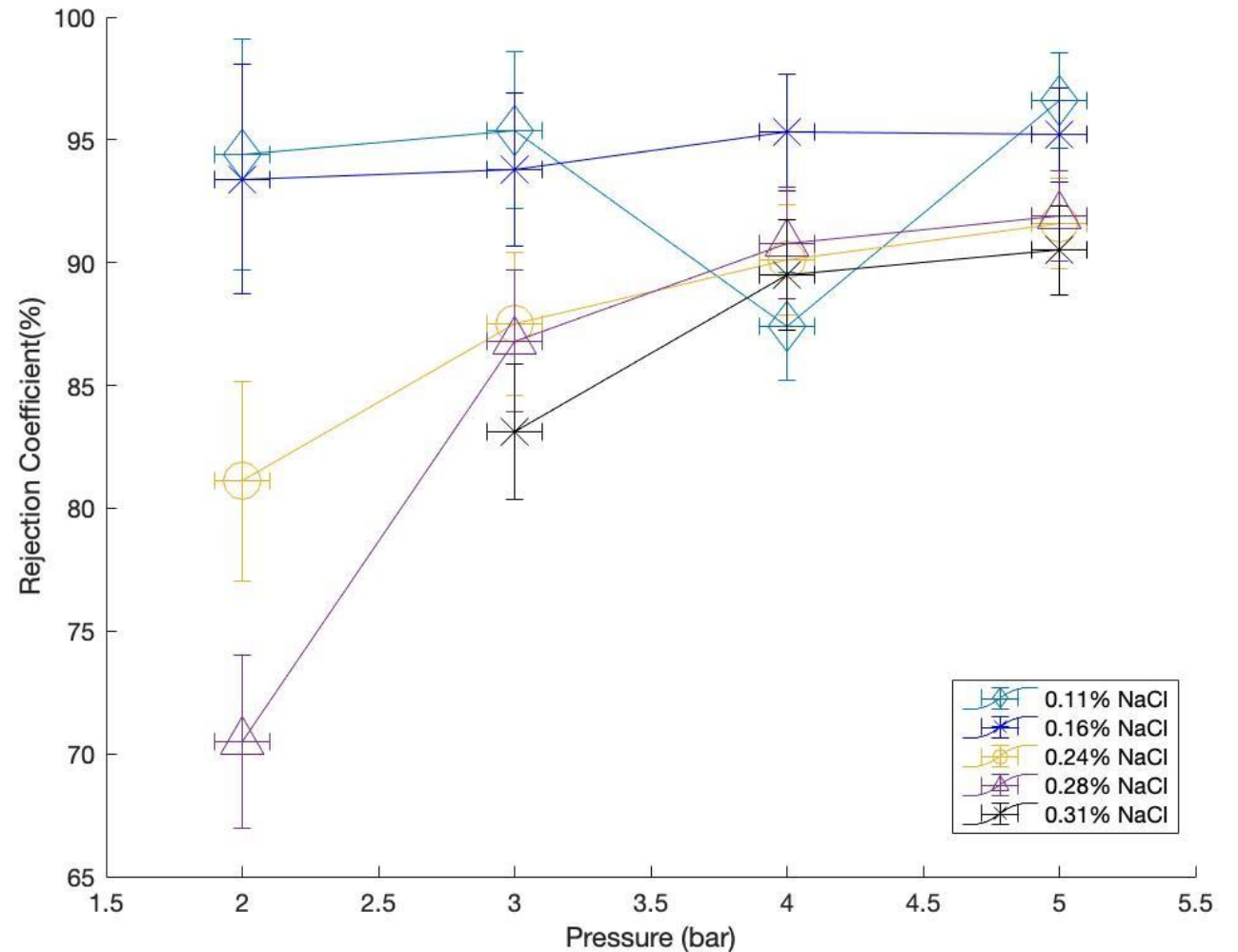


Filtration performance increases with pressure

Performance is greatly reduced below 4 bar

Rejection coefficient barely increases for 0.16% NaCl and below

Low rejection coefficient at low pressures at 0.24% NaCl and above.



Filtration performance increases with pressure

The recovery fraction increased roughly tenfold from 2% at 2 bar

We want to operate at much greater pressures than in our experiment

